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Department of Management Studies
MS 6015: Multivariate Statistical Methods for Business
Quiz 1 A
February 25, 2026

1. This quiz has a total 11 questions worth 20 marks.
2. The duration of the quiz is 50 minutes.
3. Read all questions carefully.
4. Please write your answers in the space provided
5. *All the best.*

4|5

Section 1: Multiple Choice- 5 × 1=5 marks

1. A sample is used to obtain a 95% confidence interval for the mean of a population. The confidence interval goes from 78.21 to 87.64. If the same sample had been used to test the null hypothesis that the mean of the population differs from 90, the null hypothesis could be rejected at a level of significance of 0.05.

   1 (a) True

   (b) False

2. Consider the following partial ANOVA table:

Source of Variation	SS	df	MS	F
Treatments	75	A	25	D
Error	B	C	3.75	
Total	135	19		

   1 Value of A, B, C, D are

   (a) $A = 3, B = 60, C = 16, D = 1.25$

   (b) $A = 3, B = 60, C = 16, D = 6.67$

   (c) $A = 4, B = 60, C = 15, D = 1.25$

   (d) $A = 4, B = 60, C = 15, D = 6.67$

3. If two random samples of sizes 30 and 36 are selected independently from two populations with means 78 and 85, and standard deviations 12 and 15, respectively, then the standard error of the difference $\bar{X}_1 - \bar{X}_2$ is equal to:

   1 (a) 0.904

   (b) 3.324

   (c) 3.391

   (d) 0.833

4. A professor of linguistics refutes the claim that the average student spends 3 hours studying for the midterm exam. She thinks they spend more time than that. Which hypotheses are used to test the claim?

   (a) $H_0 : \mu \neq 3$ vs. $H_1 : \mu > 3$

   (b) $H_0 : \mu = 3$ vs. $H_1 : \mu \neq 3$

   (c) $H_0 : \mu \neq 3$ vs. $H_1 : \mu = 3$

   + (d) $H_0 : \mu = 3$ vs. $H_1 : \mu < 3$

I think the options are wrong, the correct should be. $H_0 : \mu = 3$ vs $H_1 : \mu > 3$

5. Two independent samples of sizes 25 and 35 are randomly selected from two normal populations with equal variances (assumed to be unknown). In order to test the difference between the population means, the test statistic is:

1

(a) a standard normal random variable.

(b) approximately standard normal random variable.

(c) Student t-distributed with 58 degrees of freedom.

(d) Student t-distributed with 33 degrees of freedom.

Section 2: Multiple Choice- 5 × 2=10 marks. 6/10

Next two questions are based on the same description:
A wildlife ecologist measured x_1 = tail length (in millimeters) and x_2 = wing length (in millimeters) for a sample of $n = 45$ female hook-billed kites.

0 6. The sample values are $\bar{\mathbf{x}} = \begin{pmatrix} 193.62 \\ 279.78 \end{pmatrix}$. $S^{-1} = \begin{pmatrix} 0.02 & -0.01 \\ -0.01 & 0.01 \end{pmatrix}$.
The value of the test statistic to test $H_0 : \boldsymbol{\mu} = (190, 275)'$ versus $H_a : \boldsymbol{\mu} \neq (190, 275)'$. is: (Explain your answer)

$T^2 = n(\bar{x} - \mu)^T S^{-1} (\bar{x} - \mu)$

(a) 393.12 ×

(b) 5.42

(c) 0.12

(d) 5.54

7. Here are some values of the t distribution.

α	$t_{\alpha,44}$
0.05	1.6802 ←
0.01	2.4141
0.0125	2.3207
0.0025	2.9555

×

The 95% Bonferroni intervals for the means of x_1 and x_2 are. (Explain your answer)

(a) [189.42, 197.82] and [274.26, 285.30] respectively

(b) [189.82, 197.42] and [274.78, 284.77] respectively

(c) [190.32, 196.92] and [275.44,284.12] respectively ×

(d) Insufficient information to compute the intervals

Next three questions are based on the following description: A MANOVA study compares the performance of students from three different training programs on two dependent variables. Each group had 15 students. Let $\boldsymbol{\mu}_1, \boldsymbol{\mu}_2, \boldsymbol{\mu}_3$ denote the population mean vectors.

8. Which of the following represents the correct null hypothesis?

2 (a) At least one component of the mean vector is zero.

(b) $\boldsymbol{\mu}_1 = \boldsymbol{\mu}_2 = \boldsymbol{\mu}_3$

(c) All covariance matrices are identity matrices.

(d) The dependent variables are uncorrelated.

9. For the MANOVA test, Wilks' Lambda is computed as $\Lambda = 0.42$. The sampling distribution of Wilks' Lambda is given by

$$F = \left(\frac{\sum n_l - g - 1}{g - 1}\right)\left(\frac{1 - \sqrt{\Lambda}}{\sqrt{\Lambda}}\right),$$

the value of the test statistic is: (Explain your answer)

2 (a) 2.51

(b) 5.64

(c) 11.1

(d) 0.84

we put $\sum n_l = 15 + 15 + 15 = 45$
$g = 3$, and $\Lambda = 0.42$ in
the given relation

10. The critical value at $\alpha = 0.05$ is $F_{4,82} \approx 2.48$. What is the correct conclusion?

2 (a) Fail to reject H_0; all mean vectors are equal.

(b) Reject H_0; at least one population mean vector differs.

(c) Reject H_0; the covariance matrices are unequal.

(d) No decision can be made.

Section 3: Short answer question-Answer in space provided-5 marks

11 To compare the performance of large-cap and small-cap stocks, we consider the simple returns (in percentages) of CRSP decile portfolios. The large-cap stocks are represented by deciles 1 and 2, whereas the small-cap stocks are represented by deciles 9 and 10. The sample period is from 2001 to 2013 with $n_1 = n_2 = 156$. For simplicity, ignore any serial correlations in the data, if any.

Let $\boldsymbol{\mu}_\ell$ and $\boldsymbol{\Sigma}_\ell$ be the mean vector and covariance matrix of the large-cap stock returns, whereas $\boldsymbol{\mu}_s$ and $\boldsymbol{\Sigma}_s$ denote their counterparts for the small-cap stocks. Summary statistics of the data are given below.

Large Cap

$$\bar{\boldsymbol{x}}_\ell = \begin{pmatrix} 0.5 \\ 0.8 \end{pmatrix}, \qquad \boldsymbol{S}_\ell = \begin{pmatrix} 18 & 20 \\ 20 & 23 \end{pmatrix}, \qquad n_1 = 156$$

Small Cap

2

$$\bar{\boldsymbol{x}}_s = \begin{pmatrix} 1.2 \\ 1.4 \end{pmatrix}, \qquad \boldsymbol{S}_s = \begin{pmatrix} 48 & 42 \\ 42 & 49 \end{pmatrix}, \qquad n_2 = 156$$

Answer the following questions.

(a) What is the null hypothesis and alternate hypothesis. State in words and in mathematical expression. **1 mark**

- H_0: The returns of large cap and small cap funds are same.
  H_1: The returns of large cap and small cap funds are different.
- $H_0: \delta = \bar{X}_\ell - \bar{X}_s = \bar{0}$; H_1 : $\delta = (\bar{X}_\ell - \bar{X}_s) \neq \bar{0}$ → Hypothesis always on unknown $\bar{X}$ is known

(b) Compute the test statistic. Give the necessary steps involved. **2 mark**

$\boxed{T^2 = 1.196}$, $\delta = (0.7 \;\; 0.6)^T$

How??

1

$$S_{pool} = \frac{(n_1 - 1) S_1 + (n_2 - 1) S_2}{n_1 + n_2 - 2},$$

$$T^2 = (\bar{X}_1 - \bar{X}_2 - \delta_0)^T \left(\left(\frac{1}{n_1} + \frac{1}{n_2} \right) S_{pool} \right)^{-1} (\bar{X}_1 - \bar{X}_2 - \delta_0)$$

(c) If the value of the the F-critical is 3.02, what would your conclusion be? State the same with respect to the null hypothesis you have raised. **2 mark**

1

Since, $T^2 < T_{crit}^2$, we ~~conclude~~ we fail to reject the NULL Hypothesis. i.e, $\bar{X}_\ell - \bar{X}_s = 0$
In other words, The returns of large cap and small cap funds are the same.